

Beyond Bicubic: Benchmarking Single Image Super-Resolution Under Physically-Grounded Complex Degradations

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Abstract

Single Image Super-Resolution (SISR) has witnessed remarkable progress, yet the dominant evaluation paradigm relies on benchmarks with simple bicubic downsampling, failing to capture the complex degradations in real-world imaging. This mismatch leads to inflated metrics that poorly predict in-the-wild generalization.

We present REALDEG-BENCH, a large-scale benchmark accompanied by PHYSDEG, a physically-grounded, multi-stage degradation pipeline modeling the image formation process from scene radiance to stored pixel values across six stages: optical aberrations, motion blur, scene illumination, sensor noise, ISP artifacts, and lossy compression. We construct REALDEG-BENCH by applying PHYSDEG to 2,000 high-quality images spanning four semantic categories, yielding paired LR–HR data with comprehensive per-image degradation metadata.

Evaluation of four state-of-the-art methods reveals significant performance degradation: models achieving 30+ dB PSNR on Set5 attain only ~23 dB on REALDEG-BENCH, exposing a 6–8 dB generalization gap. Furthermore, models designed for blind SR (e.g., Real-ESRGAN) exhibit superior perceptual quality despite lower distortion metrics, underscoring the need for multi-faceted evaluation. Our benchmark and code will be publicly released.

1. Introduction

Single Image Super-Resolution (SISR) aims to reconstruct a high-resolution (HR) image $I_{HR} \in \mathbb{R}^{sH \times sW \times 3}$ from its low-resolution (LR) counterpart $I_{LR} \in \mathbb{R}^{H \times W \times 3}$, where s denotes the upsampling scale factor. This ill-posed inverse problem has broad applicability in medical imaging [24], remote sensing [23], surveillance [37], and consumer photography. The field has progressed from dictionary-based methods [32] through CNNs [9, 18] to Vision Transformers [8, 17] and GANs [27, 28], with top models exceeding 32 dB PSNR on Set5 [2] at $\times 4$.

However, the dominant benchmarking paradigm generates LR images through a single bicubic downsampling: $I_{LR} = I_{HR} \downarrow_s^{\text{bicubic}}$. This assumes an idealized pipeline where the sole information loss is spatial resolution reduction via a known kernel. In practice, real cameras introduce optical aberrations, sensor noise, motion blur, ISP artifacts, and lossy compression [19, 36]. Models trained under the bicubic assumption often fail catastrophically on these real-world degradations [28], a phenomenon we term the *degradation gap*.

In this paper, we address this gap through two contributions. First, we introduce PHYSDEG, a *physically-grounded, multi-stage degradation pipeline* organizing ten distinct degradation operations into six ordered stages—optical, motion, scene, sensor, ISP, and compression—each parameterized by quantities with clear physical interpretations. Second, we present REALDEG-BENCH, a large-scale benchmark of 2,000 diverse image pairs across four semantic categories (urban architecture, natural landscapes, portraits, macro objects) with comprehensive per-image degradation metadata.

Our evaluation of four SOTA methods reveals:

- Substantial generalization gap:** models achieving >30 dB on Set5 attain only ~23–24 dB on REALDEG-BENCH (6–8 dB drop).
- Perception–distortion divergence:** Real-ESRGAN obtains the best LPIPS (0.444 vs. >0.55 for non-blind methods) despite the lowest PSNR, showing that distortion metrics alone are insufficient under complex degradations.
- Content-dependent vulnerability:** urban architecture is most challenging due to geometric-degradation interactions; macro objects are comparatively resilient.

Our contributions can be summarized as:

- We propose PHYSDEG, a physically-grounded degradation pipeline comprising six ordered stages with ten degradation types, each governed by physically meaningful parameters.
- We construct REALDEG-BENCH, a benchmark of 2,000 image pairs spanning four semantic categories with com-

Physically-Grounded Image Degradation Pipeline

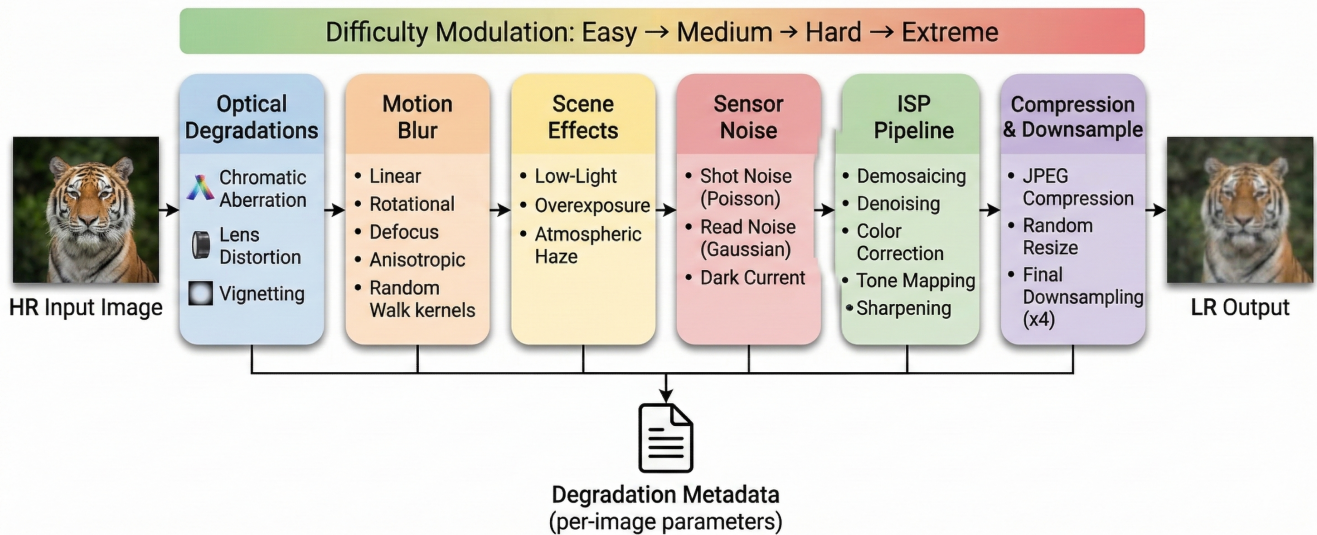


Figure 1. **Overview of the PHYSDEG degradation pipeline.** Six sequential stages model the image formation process from scene to stored file: (1) optical aberrations, (2) motion blur (5 kernel types), (3) scene illumination effects, (4) sensor noise, (5) ISP processing, and (6) compression and downsampling. Each stage is parameterized by physically meaningful quantities and applied stochastically, yielding diverse LR–HR pairs with complete degradation metadata.

plete per-image degradation metadata.

- We quantify the degradation gap: SOTA methods suffer a 6–8 dB PSNR drop from bicubic to realistic degradations, motivating future research in degradation-robust SR.

2. Related Work

Single image super-resolution. Deep SISR has evolved rapidly from SRCNN [9] through residual architectures (EDSR [18], RCAN [39]) to Transformer-based methods. SwinIR [17] adapted shifted-window attention for efficient image restoration, and subsequent works—HAT [8], DSCP-SR [31], SeemoRe [33]—have further pushed quantitative benchmarks. In parallel, GAN-based approaches (SR-GAN [16], ESRGAN [27]) demonstrated that perceptual and adversarial losses produce photo-realistic textures, establishing a fundamental tension between distortion and perceptual quality [3]. Recent perceptual objectives include self-supervised DINO features [6] and frequency-domain losses [11].

Degradation modeling. The classical degradation model $I_{LR} = (I_{HR} \otimes k) \downarrow_s + n$ [9] is typically simplified to bicubic downsampling with $n=0$. Blind SR methods address more complex scenarios: IKC [12] iteratively corrects kernels, BSRGAN [36] randomly shuffles blur-downsample-noise operations, and Real-ESRGAN [28] applies a second-order

degradation pipeline with sinc filters. Despite advances in degradation-aware *training*, the *evaluation* paradigm has not kept pace—models are still predominantly assessed on bicubically degraded benchmarks. Our work draws on camera ISP modeling [34] and sensor noise characterization [10, 13] to construct a physically principled pipeline.

SISR benchmark datasets. The widely used benchmarks—Set5 [2], Set14 [35], BSD100 [20], Urban100 [14]—all employ bicubic downsampling. Real-world alternatives like RealSR [5] (595 pairs, single camera) and DRealSR [30] (840 pairs, five cameras) provide genuine degradations but are limited in scale and diversity, with potential alignment artifacts. Our REALDEG-BENCH occupies a distinct niche: unlike bicubic benchmarks, we model diverse, physically-grounded degradations; unlike real-captured datasets, our synthetic approach provides exact ground truth without alignment errors. Uniquely, REALDEG-BENCH includes per-image degradation metadata enabling fine-grained robustness analysis (Table 1).

3. Proposed Degradation Pipeline

We introduce PHYSDEG, a physically-grounded degradation pipeline that models the sequential transformation of scene radiance into a stored low-resolution image. Unlike exist-

Table 1. **Comparison of SISR benchmark datasets.** REALDEG-BENCH uniquely combines large scale, diverse categories, complex degradations, and per-image metadata.

Dataset	Size	Degrad.	Categories	Metadata	GT
Set5 [2]	5	Bicubic	×	×	✓
Set14 [35]	14	Bicubic	×	×	✓
BSD100 [20]	100	Bicubic	×	×	✓
Urban100 [14]	100	Bicubic	1	×	✓
DIV2K [1]	1,000	Bicubic	×	×	✓
RealSR [5]	595	Real (1 cam.)	×	×	✓
DRealSR [30]	840	Real (5 cam.)	×	×	~
REALDEG-BENCH	2,000	Complex	4	✓	✓

ing approaches that apply degradations in a flat or shuffled manner [28, 36], our pipeline organizes degradation operations into six physically-ordered stages that mirror the actual image formation process:

$$I_{LR} = (\mathcal{D}_{\text{comp}} \circ \mathcal{D}_{\text{isp}} \circ \mathcal{D}_{\text{sensor}} \circ \mathcal{D}_{\text{scene}} \circ \mathcal{D}_{\text{motion}} \circ \mathcal{D}_{\text{optical}})(I_{HR}) \downarrow_{s, \gamma}$$

where each operator $\mathcal{D}_{(\cdot)}$ is parameterized by physically meaningful quantities and applied stochastically according to a difficulty-modulated probability.

3.1. Stage 1: Optical Degradations

Optical degradations originate from imperfections in the lens system. We model three effects: *chromatic aberration* via wavelength-dependent channel shifts $\delta_c \sim \mathcal{U}(-2.5, 2.5)$ pixels [15]; *radial lens distortion* using the Brown–Conrady model [4]:

$$r' = r(1 + k_1 r^2), \quad k_1 \sim \mathcal{U}(-0.15, 0.15); \quad (2)$$

and *vignetting* via the $\cos^4$ fall-off approximation $V(x, y) = 1 - s \cdot (d(x, y)/d_{\text{max}})^2$ with $s \sim \mathcal{U}(0.1, 0.5)$.

3.2. Stage 2: Motion Blur

Motion blur arises during the exposure interval and is modeled as $I_{\text{blur}} = I * k_{\theta}$. We implement five physically-motivated kernel types to capture the diversity of real-world motion: (1) *linear* uniform kernels along direction ϕ with length $l \sim \mathcal{U}(5, 20)$ px; (2) *defocus* disk kernels of radius $r \sim \mathcal{U}(2, 10)$ px, approximating circular-aperture bokeh [22]; (3) *anisotropic Gaussian* kernels with independent scales $\sigma_x, \sigma_y \sim \mathcal{U}(0.5, 4.0)$ and rotation $\phi \sim \mathcal{U}(0, \pi)$; (4) *rotational* blur simulating in-plane camera rotation; and (5) *random walk* trajectories $\mathbf{p}_t = \sum_{\tau=1}^t \epsilon_{\tau}$ with $\epsilon_{\tau} \sim \mathcal{N}(0, \sigma_{\text{step}}^2 \mathbf{I}_2)$ simulating camera shake.

3.3. Stage 3: Scene Illumination Effects

We model two scene-level effects. *Low-light degradation* applies gamma correction with noise amplification: $I_{\text{low}} = I^{\gamma} + \mathcal{N}(0, \sigma_0^2 b^2)$ where $\gamma \sim \mathcal{U}(1.5, 3.0)$ and $b \sim \mathcal{U}(1.2, 2.0)$, modeling the increased noise visibility in

underexposed imagery [7]. *Atmospheric scattering* follows the Koschmieder model [21]: $I_{\text{haze}} = t \cdot J + A(1 - t)$, where $A \sim \mathcal{U}(0.8, 0.95)$ is the atmospheric light and $t = \exp(-\beta \cdot d)$ with $\beta \sim \mathcal{U}(0.1, 0.6)$.

3.4. Stage 4: Sensor Noise

Camera sensor noise is a composite of multiple physical sources [10, 13]. We model three components: (1) signal-dependent *shot noise* (Poisson):

$$I_{\text{shot}}(x, c) = \text{Poisson}\left(\frac{I(x, c)}{\alpha_c}\right) \cdot \alpha_c, \quad (3)$$

where $\alpha_c = \alpha \cdot \lambda_c$ with $\alpha \sim \mathcal{U}(0.01, 0.08)$ and channel factors $[\lambda_R, \lambda_G, \lambda_B] = [1.1, 1.0, 1.05]$; (2) spatially-correlated *read noise* (Gaussian) with $\sigma_r \sim \mathcal{U}(0.005, 0.025)$; and (3) *dark current* offset $d \sim \mathcal{U}(0.001, 0.012)$. The composite model is $I_{\text{noisy}} = \text{clip}[\text{Shot}(I) + \text{Read}(I) + d, 0, 1]$.

3.5. Stage 5: Image Signal Processing (ISP)

Consumer cameras apply an ISP pipeline that introduces its own artifacts. We model five sub-stages in their canonical processing order: $\mathcal{D}_{\text{isp}} = \mathcal{S} \circ \mathcal{T} \circ \mathcal{C} \circ \mathcal{F} \circ \mathcal{M}$, where $\mathcal{M}$ denotes demosaicing (cross-channel leakage simulation), $\mathcal{F}$ bilateral denoising [25], $\mathcal{C}$ stochastic color correction via a perturbed 3×3 matrix with saturation adjustment, $\mathcal{T}$ gamma-based tone mapping with highlight compression, and $\mathcal{S}$ unsharp masking.

3.6. Stage 6: Compression and Downsampling

The final stage applies JPEG compression ($q \sim \mathcal{U}(25, 85)$), an optional random intermediate resize ($f \sim \mathcal{U}(0.7, 1.3)$) simulating social media processing, and terminal downsampling by factor s using a randomly selected kernel from {bicubic, bilinear, Lanczos}.

3.7. Difficulty Modulation

Each degradation is applied independently with a difficulty-modulated probability. We define four levels—*easy*, *medium*, *hard*, *extreme*—controlling the base application probability p and maximum concurrent degradations N_{max} : $(p, N_{\text{max}}) \in \{(0.3, 2), (0.5, 3), (0.7, 5), (0.9, 7)\}$. Stage-specific modifiers further refine rates: optical effects are scaled by $0.7 \times$ (less common), while ISP and sensor noise are scaled by $2 \times$ and $1.1 \times$ (highly prevalent). Table 2 summarizes all degradation types and their parameters.

4. REALDEG-BENCH: Dataset Construction

4.1. Source Image Collection

We curate 2,000 high-quality HR images from Pexels¹ (CC0 license), organized into four semantically distinct categories:

¹<https://www.pexels.com>

Table 2. **Summary of degradation types in PHYSDEG.** Application probabilities shown for the default random difficulty mode.

Stage	Degradation	Key Parameters	Prob.
Optical	Chromatic aberr.	$\delta \in [-2.5, 2.5]$ px	~ 0.4
	Lens distortion	$k_1 \in [-0.15, 0.15]$	~ 0.4
	Vignetting	$s \in [0.1, 0.5]$	~ 0.3
Motion	Motion blur	5 kernel types	0.50
Scene	Low-light	$\gamma \in [1.5, 3.0]$	~ 0.4
	Atmospheric	$\beta \in [0.1, 0.6]$	~ 0.3
Sensor	Composite noise	α, σ_r, d	~ 0.9
ISP	5-stage ISP	See Sec. 3.5	~ 0.8
Comp.	JPEG compress.	$q \in [25, 85]$	0.70
	Random resize	$f \in [0.7, 1.3]$	0.60

Urban Architecture (~ 500): buildings and cityscapes with geometric patterns and fine structural details; **Natural Landscapes** (~ 500): outdoor scenes with organic textures and complex foliage; **Portraits** (~ 500): human subjects with smooth skin gradients and fine hair details; **Macro Objects** (~ 500): close-up photography with fine-grained textures. All sources satisfy a minimum resolution of 1024×1024 pixels.

4.2. Generation Process

Algorithm 1 outlines the procedure. For each HR image, we apply the PHYSDEG pipeline (Sec. 3) in random difficulty mode ($p \sim \mathcal{U}(0.4, 0.8)$, $N_{\max} \sim \mathcal{U}(2, 6)$), then downsample by $s=4$. Complete degradation metadata $\mathbf{m}_i$ is recorded for every pair—the exact sequence of applied degradations, their parameters, and difficulty configuration—enabling stratified evaluation, degradation-conditioned model training, and detailed failure analysis.

4.3. Dataset Statistics

Table 3 summarizes the key statistics. On average, each image undergoes 3.2 ± 1.4 distinct degradation operations, with 15–40 parameters recorded per image. The stochastic nature of PHYSDEG ensures broad coverage of the degradation parameter space: JPEG quality factors span 25–85, sensor noise levels range over an order of magnitude, and all five motion blur types are represented.

Algorithm 1 Dataset generation procedure

Require: Source images $\{I_i^{HR}\}_{i=1}^N$, scale factor s , pipeline $\mathcal{P}$

Ensure: Paired dataset $\{(I_i^{LR}, I_i^{HR}, \mathbf{m}_i)\}_{i=1}^N$

```

1: for  $i = 1$  to  $N$  do
2:   Sample difficulty:  $p \sim \mathcal{U}(0.4, 0.8)$ ,  $N_{\max} \sim \mathcal{U}(2, 6)$ 
3:   Initialize metadata:  $\mathbf{m}_i \leftarrow \emptyset$ 
4:   for each stage  $\mathcal{D}_k$  in  $\mathcal{P}$  do
5:     Sample application:  $u \sim \mathcal{U}(0, 1)$ 
6:     if  $u < p \cdot w_k$  and  $|\mathbf{m}_i| < N_{\max}$  then
7:        $I_i^{\text{deg}}, \theta_k \leftarrow \mathcal{D}_k(I_i^{\text{deg}}; \text{random})$ 
8:        $\mathbf{m}_i \leftarrow \mathbf{m}_i \cup \{(k, \theta_k)\}$ 
9:     end if
10:  end for
11:   $I_i^{LR} \leftarrow \text{JPEG}(I_i^{\text{deg}}; q) \downarrow_{\phi, s}$ 
12: end for

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Table 3. **Dataset statistics of REALDEG-BENCH.**

Statistic	Urban	Landscape	Portrait	Macro
Number of pairs	~500	~500	~500	~500
Min HR resolution	1024 × 1024 pixels			
Scale factor	×4			
<i>Degradation coverage (mean applied per image):</i>				
Optical degradations	0.8–1.2 per image			
Motion blur	0.3–0.5 per image			
Sensor noise	0.7–0.9 per image			
ISP artifacts	2.5–4.0 stages per image			
JPEG compression	0.65–0.75 per image			
Avg. degradations/image	3.2 ± 1.4			
Difficulty range	Easy – Extreme (continuous)			
Metadata fields/image	15–40 parameters			

5. Experiments

We validate REALDEG-BENCH through a comprehensive evaluation of state-of-the-art SISR methods, demonstrating the degradation gap between standard benchmarks and our realistic degradation protocol.

5.1. Setup

Evaluated methods. We select four representative models: **DSCF-SR** [31] (Transformer with dynamic sparse cross-attention), **SeemoRe-B** and **SeemoRe-T** [33] (mixture-of-experts, Base and Tiny variants), and **Real-ESRGAN** [28] (GAN-based blind SR with second-order degradation training). The first three are trained under bicubic degradation on DIV2K/Flickr2K; Real-ESRGAN uses its own stochastic degradation pipeline. All models use officially released $\times 4$ weights; inference is on an NVIDIA Tesla P100 GPU.

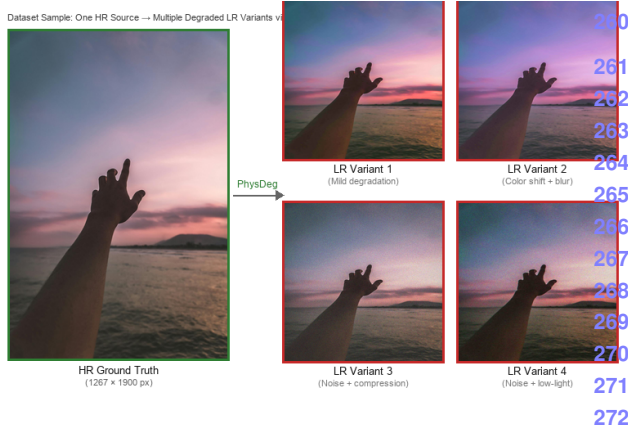


Figure 2. **Example LR-HR pairs from REALDEG-BENCH.** Left: HR ground truth (1024×1024). Right: four LR variants (256×256) produced by PHYSDEG at different difficulty levels, exhibiting realistic combinations of blur, noise, color shifts, and compression artifacts.

Table 4. **Overall performance on REALDEG-BENCH ($\times 4$).** Non-blind methods cluster in distortion metrics; the blind method (Real-ESRGAN) achieves lower PSNR but substantially better perceptual quality.

Method	Training	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
DSCF-SR [31]	Bicubic	23.82	0.674	0.556
SeemoRe-B [33]	Bicubic	23.86	0.673	0.564
SeemoRe-T [33]	Bicubic	23.85	0.673	0.565
Real-ESRGAN [28]	Blind deg.	22.65	0.686	0.444

Metrics. We employ PSNR and SSIM [29] (computed on the Y channel) as distortion metrics, and LPIPS [38] as a learned perceptual quality metric.

5.2. Overall Results

Table 4 presents aggregate results across all 1,140 test pairs. The best non-blind methods achieve only ~ 23.9 dB PSNR on REALDEG-BENCH, compared to > 30 dB on Set5 under bicubic degradation—a **6–8 dB gap** that directly quantifies the degradation gap motivating our work. DSCF-SR, SeemoRe-B, and SeemoRe-T converge to nearly identical performance (within 0.04 dB), suggesting that under complex degradations the performance ceiling is determined by the degradation mismatch rather than architectural capacity.

Real-ESRGAN presents a striking counterpoint: while achieving the lowest PSNR (-1.21 dB below non-blind methods), it obtains the best SSIM (0.686) and substantially better LPIPS (0.444 vs. > 0.555 , a 20% relative improvement). This perception-distortion divergence [3] underscores the need for multi-metric evaluation under complex degradations.

5.3. Category-Wise Analysis

Table 5 reveals content-dependent vulnerability patterns. **Urban architecture** is consistently the most challenging category (~ 23.3 dB for non-blind methods vs. ~ 24.5 dB for portraits/macro), as geometric structures interact adversely with blur and compression artifacts. **Macro objects** are comparatively resilient due to textural regularity and self-similar structures. Real-ESRGAN’s LPIPS advantage is universal (0.119–0.131 improvement), largest for natural landscapes where adversarially generated organic textures are perceptually more convincing. In portraits, the PSNR gap narrows (-1.39 dB) while the LPIPS gap widens, highlighting the acute perception-distortion tradeoff for facial content.

5.4. Standard Benchmarks vs. REALDEG-BENCH

Table 6 contextualizes the degradation gap. Even Real-ESRGAN, explicitly trained for blind SR, drops ~ 5 dB relative to Urban100, indicating that degradation-aware training has substantial room for improvement against the diversity of PHYSDEG. The high per-image variance (5.7–7.1 dB PSNR standard deviation) further highlights that model performance on realistic data is highly image-dependent, motivating per-image metadata for fine-grained analysis.

5.5. Qualitative Results

Figure 3 shows representative failure cases. Non-blind methods exhibit characteristic artifacts: noise amplification, color fringing from chromatic aberration, and ringing where the assumed bicubic kernel differs from the actual degradation. Real-ESRGAN produces perceptually more coherent outputs but occasionally hallucinates textures not present in the ground truth.

5.6. Toward Degradation-Aware SR

The per-image degradation metadata in REALDEG-BENCH enables degradation-conditioned SR. As a proof of concept, we describe a baseline Degradation-Aware Swin Transformer (DAST) that predicts a degradation vector from the LR input and conditions the backbone via SFT layers [26]: $\mathbf{F} = (\gamma(\mathbf{d}) + 1) \odot \mathbf{F} + \beta(\mathbf{d})$. The estimator is supervised using ground-truth metadata, enabling precise characterization without iterative refinement—a capability impossible to develop with existing benchmarks.

6. Conclusion

We have presented REALDEG-BENCH, a benchmark for evaluating SISR under realistic, physically-grounded degradations, accompanied by PHYSDEG—a multi-stage degradation pipeline modeling the image formation process from optical aberrations to lossy compression. Our benchmarking of four SOTA methods reveals a 6–8 dB PSNR gap between standard bicubic benchmarks and REALDEG-BENCH, with

Table 5. **Category-wise performance on REALDEG-BENCH ($\times 4$).** Best per metric and category in **bold**. Urban architecture is most challenging; Real-ESRGAN leads LPIPS across all categories.

Method	Urban Architecture			Natural Landscapes			Portraits & People			Macro Objects		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
DSCF-SR	23.26	0.692	0.505	23.12	0.676	0.539	24.55	0.658	0.602	24.44	0.669	0.581
SeemoRe-B	23.35	0.693	0.511	23.14	0.674	0.545	24.58	0.657	0.607	24.48	0.667	0.584
SeemoRe-T	23.34	0.692	0.510	23.13	0.674	0.544	24.57	0.657	0.606	24.48	0.667	0.584
Real-ESRGAN	21.89	0.702	0.383	22.17	0.701	0.408	23.16	0.643	0.532	23.43	0.694	0.463

Table 6. **Performance gap: bicubic benchmarks vs. REALDEG-BENCH.** The 6–8 dB drop reveals a severe generalization gap.

Method	Set5	Set14	BSD100	Urban100	REALDEG-BENCH
Non-blind SOTA	~33	~29	~28	~27	~23.9
Real-ESRGAN	~28	~25	~25	~24	22.7
Δ (drop)	–	–	–	–	6–8 dB

non-blind methods converging to similar performance regardless of architecture, and a striking perception–distortion divergence for the blind method Real-ESRGAN.

The per-image degradation metadata—unique to REALDEG-BENCH—opens several research directions: degradation-conditioned models that adapt to input-specific degradation profiles (*cf.* Sec. 5.6), stratified robustness evaluation by degradation type and severity, and curriculum learning with calibrated difficulty levels. The modular design of PHYSDEG allows extension with new degradation types or adaptation to other restoration tasks. We will publicly release the complete dataset, pipeline code, and evaluation scripts to facilitate reproducible research.

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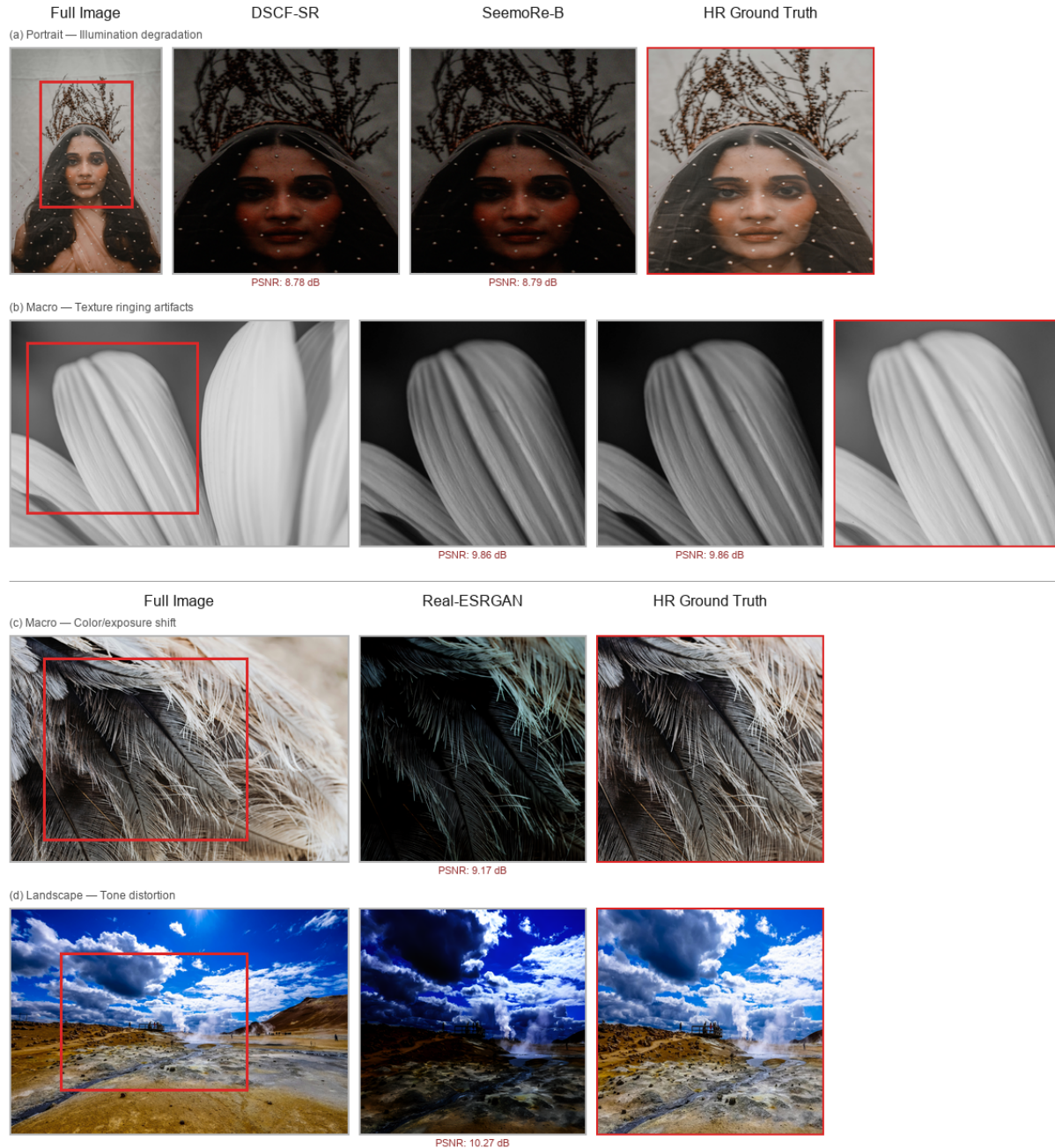


Figure 3. **Qualitative comparison on worst-case samples from REALDEG-BENCH. Top (a–b):** Non-blind methods fail catastrophically on realistic degradations—severe illumination causes darkened outputs (PSNR <9 dB), and texture ringing emerges on fine structures. **Bottom (c–d):** Real-ESRGAN struggles with extreme color/exposure shifts. These failures are invisible in standard bicubic benchmarks.

References

- [1] Eirikur Agustsson and Radu Timofte. NTIRE 2017 challenge on single image super-resolution: Dataset and study. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recog. Workshops*, pages 126–135, 2017.
- [2] Marco Bevilacqua, Aline Roumy, Christine Guillemot, and Marie-Line Alberi-Morel. Low-complexity single-image super-resolution based on nonnegative neighbor embedding. In *Proc. Brit. Mach. Vis. Conf.*, 2012.
- [3] Yochai Blau and Tomer Michaeli. The perception-distortion tradeoff. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recog.*, pages 6228–6237, 2018.
- [4] Duane C Brown. Close-range camera calibration. Technical

- Report 8, 1971.
- [5] Jianrui Cai, Hui Zeng, Hongwei Yong, Zisheng Cao, and Zhang. Toward real-world single image super-resolution: a new benchmark and a new model. In *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, pages 3086–3095, 2019.
- [6] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, pages 9650–9660, 2021.
- [7] Chen Chen, Qifeng Chen, Jia Xu, and Vladlen Koltun. Learning to see in the dark. In *Proc. IEEE/CVF Conf. Comput. Pattern Recog.*, pages 3291–3300, 2018.
- [8] Xiangyu Chen, Xintao Wang, Jianqi Zhou, Yu Qiao, and Chao Dong. HAT: Hybrid attention transformer for image restoration. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recog.*, pages 12592–12601, 2023.
- [9] Chao Dong, Chen Change Loy, Kaiming He, and Xiaoou Tang. Image super-resolution using deep convolutional networks. *IEEE Trans. Pattern Anal. Mach. Intell.*, 38(2):2959–307, 2016.
- [10] Alessandro Foi, Mejdî Trimeche, Vladimir Katkovnik, and Karen Egiazarian. Practical poissonian-Gaussian noise modeling and fitting for single-image raw-data. *IEEE Trans. Image Process.*, 17(10):1737–1754, 2008.
- [11] Dario Fuoli, Luc Van Gool, and Radu Timofte. Fourier space losses for efficient perceptual image super-resolution. In *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, pages 2360–2369, 2021.
- [12] Jinjin Gu, Hannan Lu, Wangmeng Zuo, and Chao Dong. Blind super-resolution with iterative kernel correction. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recog.*, pages 1604–1613, 2019.
- [13] Glenn E Healey and Raghava Kondepudy. Radiometric camera calibration and noise estimation. *IEEE Trans. Pattern Anal. Mach. Intell.*, 16(3):267–276, 1994.
- [14] Jia-Bin Huang, Abhishek Singh, and Narendra Ahuja. Single image super-resolution from transformed self-exemplars. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recog.*, pages 5197–5206, 2015.
- [15] Sing Bing Kang. Automatic removal of chromatic aberration from a single image. *Proc. IEEE/CVF Conf. Comput. Pattern Recog.*, pages 1–8, 2007.
- [16] Christian Ledig, Lucas Theis, Ferenc Huszár, Jose Caballero, Andrew Cunningham, Alejandro Acosta, Andrew Aitken, Alykhan Tejani, Johannes Totz, Zehan Wang, et al. Photo-realistic single image super-resolution using a generative adversarial network. In *Proc. IEEE/CVF Conf. Comput. Pattern Recog.*, pages 4681–4690, 2017.
- [17] Jingyun Liang, Jiezhang Cao, Guolei Sun, Kai Zhang, Luc Van Gool, and Radu Timofte. SwinIR: Image restoration using swin transformer. In *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, pages 1833–1844, 2021.
- [18] Bee Lim, Sanghyun Son, Heewon Kim, Seungjun Nah, and Kyoung Mu Lee. Enhanced deep residual networks for single image super-resolution. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recog. Workshops*, pages 136–144, 2017.
- [19] Anran Liu, Yihao Liu, Jinjin Gu, Yu Qiao, and Chao Dong. Blind image super-resolution: A survey and beyond. In *IEEE Trans. Pattern Anal. Mach. Intell.*, pages 7536–7561, 2022.
- [20] David Martin, Charless Fowlkes, Doron Tal, and Jitendra Malik. A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics. In *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, pages 416–423, 2001.
- [21] Earl J McCartney. *Optics of the Atmosphere: Scattering by Molecules and Particles*. John Wiley & Sons, 1976.
- [22] Michael Potmesil and Indranil Chakravarty. A lens and aperture camera model for synthetic image generation. In *Proc. ACM SIGGRAPH*, pages 297–305, 1981.
- [23] Jacob Shermeyer and Adam Van Etten. The effects of super-resolution on object detection performance in satellite imagery. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recog. Workshops*, pages 0–0, 2019.
- [24] Wenzhe Shi, Jose Caballero, Christian Ledig, Xiaohai Zhuang, Wenjia Bai, Kanwal Bhatia, Antonio M Simoes Monteiro de Marvao, Tim Dawes, Declan O’Regan, and Daniel Rueckert. Cardiac image super-resolution with global correspondence using multi-atlas patchmatch. *Med. Image Comput. Comput. Assist. Interv.*, pages 9–16, 2013.
- [25] Carlo Tomasi and Roberto Manduchi. Bilateral filtering for gray and color images. In *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, pages 839–846, 1998.
- [26] Xintao Wang, Ke Yu, Chao Dong, and Chen Change Loy. Recovering realistic texture in image super-resolution by deep spatial feature transform. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recog.*, pages 606–615, 2018.
- [27] Xintao Wang, Ke Yu, Shixiang Wu, Jinjin Gu, Yihao Liu, Chao Dong, Yu Qiao, and Chen Change Loy. ESRGAN: Enhanced super-resolution generative adversarial networks. In *Proc. Eur. Conf. Comput. Vis.*, pages 63–79, 2018.
- [28] Xintao Wang, Liangbin Xie, Chao Dong, and Ying Shan. Real-ESRGAN: Training real-world blind super-resolution with pure synthetic data. In *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, pages 1905–1914, 2021.
- [29] Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Trans. Image Process.*, 13(4): 600–612, 2004.
- [30] Pengxu Wei, Ziwei Xie, Hannan Lu, Zongyuan Zhan, Qixiang Liu, Wangmeng Zuo, and Liang Lin. Component divide-and-conquer for real-world image super-resolution. In *Proc. Eur. Conf. Comput. Vis.*, pages 101–117, 2020.
- [31] Zhen Xiao et al. DSCF: Dynamic sparse cross-attention and frequency fusion for single image super-resolution. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recog.*, 2024.
- [32] Jianchao Yang, John Wright, Thomas S Huang, and Yi Ma. Image super-resolution via sparse representation. *IEEE Trans. Image Process.*, 19(11):2861–2873, 2010.
- [33] Eduard Zamfir et al. SeemoRe: A simple yet effective approach for efficient super-resolution via expert routing. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recog. Workshops*, 2024.
- [34] Syed Waqas Zamir, Aditya Arora, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, Ming-Hsuan Yang, and Ling Shao. CycleISP: Real image restoration via improved data synthesis. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recog.*, pages 2696–2705, 2020.

- [35] Roman Zeyde, Michael Elad, and Matan Protter. On single image scale-up using sparse-representations. In *Proc. Int. Conf. Curves Surfaces*, pages 711–730, 2012.
- [36] Kai Zhang, Jingyun Liang, Luc Van Gool, and Radu Timofte. Designing a practical degradation model for deep blind image super-resolution. In *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, pages 4791–4800, 2021.
- [37] Liangpei Zhang, Hongyan Zhang, Huanfeng Shen, and Pingxiang Li. Super-resolution reconstruction for multi-angle remote sensing images considering resolution differences. *Remote Sens.*, 2(6):1532–1554, 2010.
- [38] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recog.*, pages 586–595, 2018.
- [39] Yulun Zhang, Kunpeng Li, Kai Li, Lichen Wang, Bineng Zhong, and Yun Fu. Image super-resolution using very deep residual channel attention networks. In *Proc. Eur. Conf. Comput. Vis.*, pages 286–301, 2018.